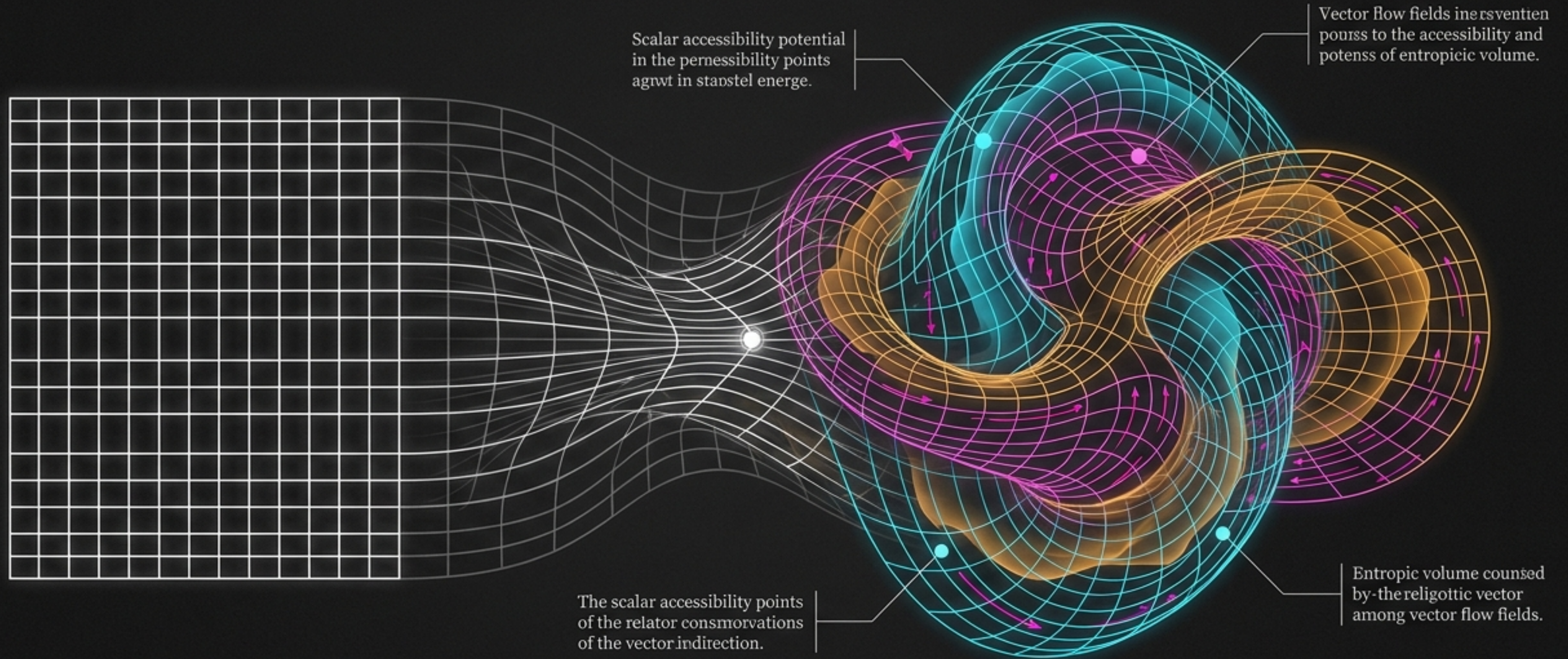


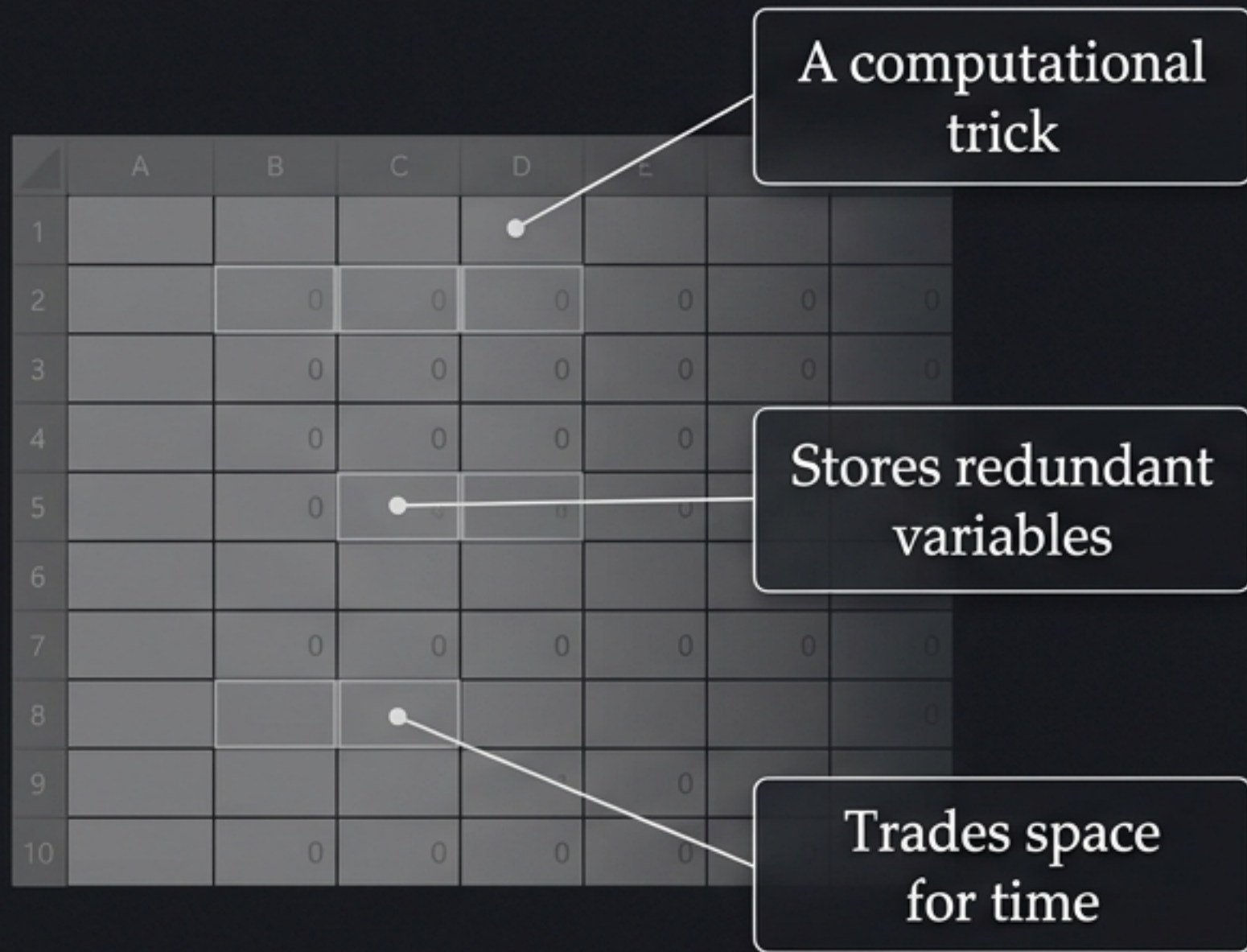
The Hidden Geometry of Optimization

Dynamic Programming as Field Theory in the Relativistic Scalar-Vector Plenum

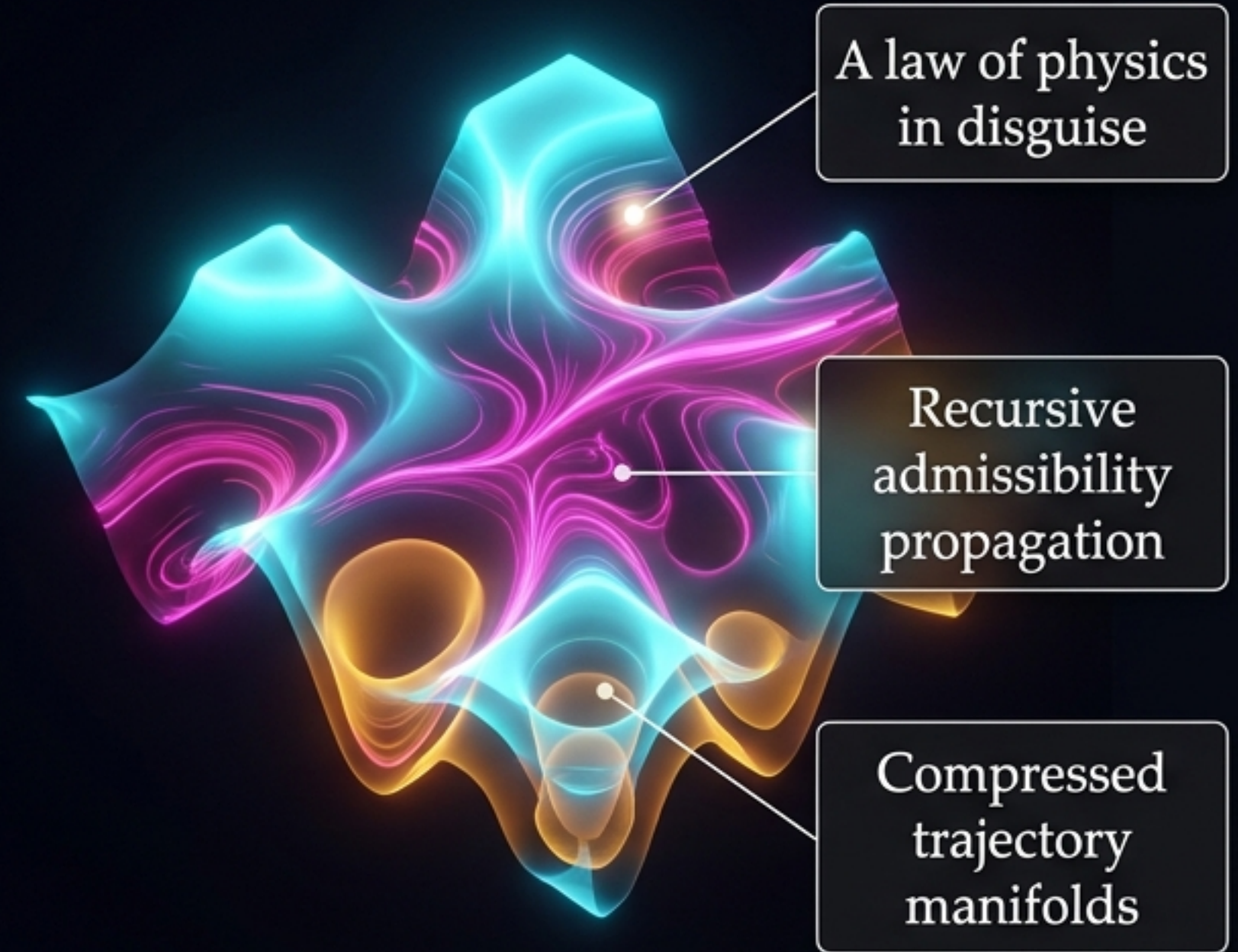


A synthesis of computational algorithms and continuous physical manifolds.

The Computational Illusion



The Ontological Reality

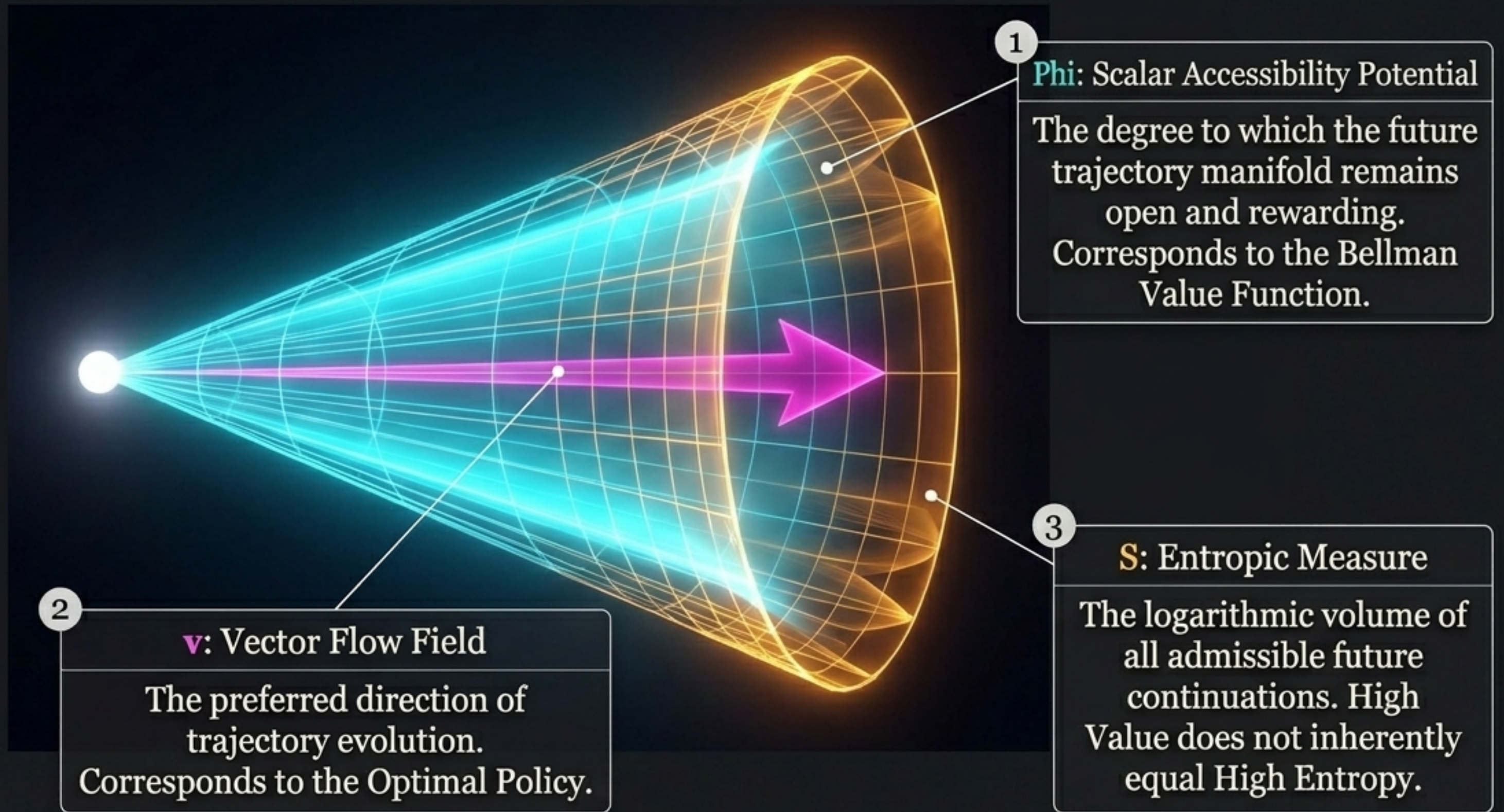


Dynamic programming does not merely compute optimal solutions; it maps the fundamental field theory of constrained future trajectories.

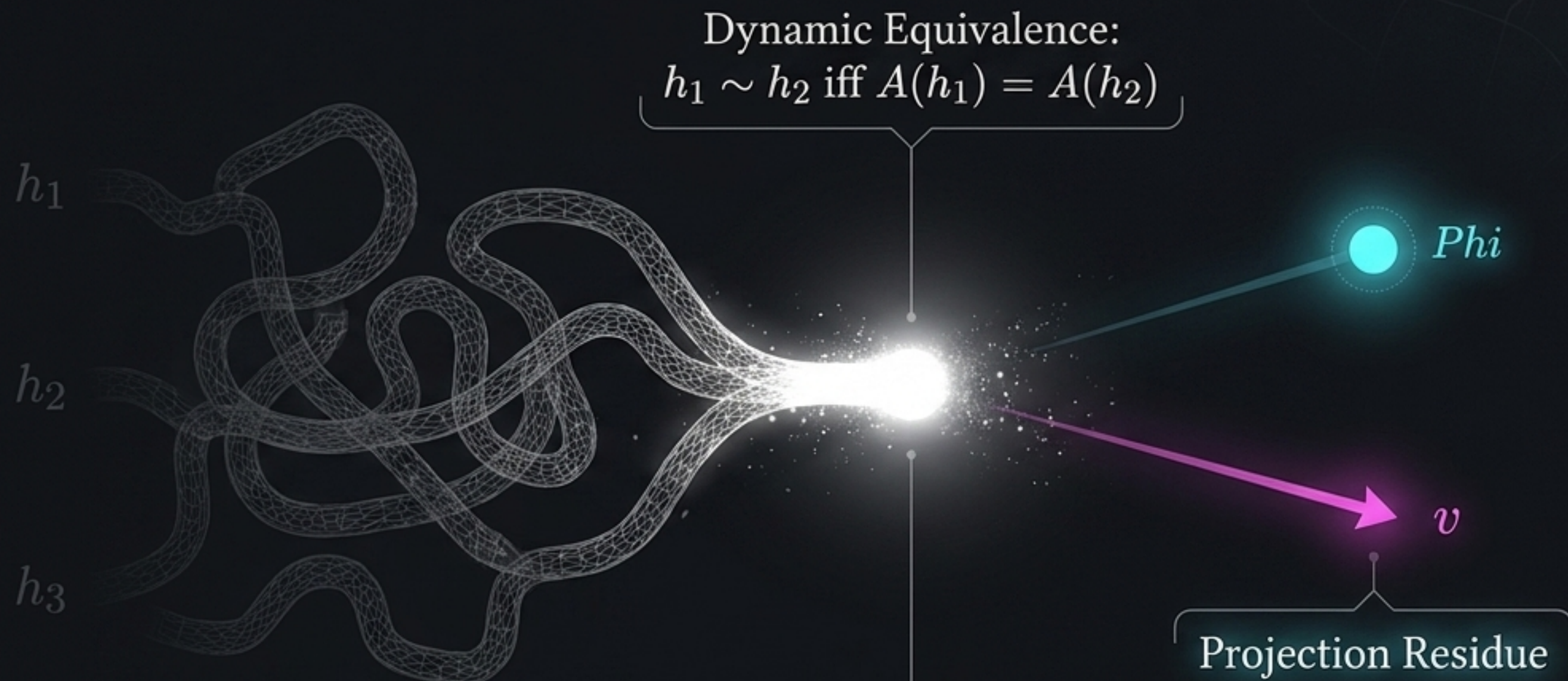
The Fundamental Identification

Entity	Classical DP View	Continuous View	RSVP Geometric View
State	Discrete Grid Node	Continuous State-Time Point	Admissibility Region (Equivalence Class)
Value	Value Function $V(s)$	Cost-to-Go $J^*(\mathbf{x}, t)$	Scalar Accessibility Potential (Φ)
Action	Optimal Policy $\pi^*(s)$	Optimal Control $u^*(\mathbf{x}, t)$	Vector Flow Field ($\mathbf{v}$)
Scope	Overlapping Subproblems	Trajectory Intersections	Logarithmic Admissibility Volume (S)
Mechanism	Bellman Update	HJB Equation	Admissibility Propagation

THE ADMISSIBILITY CONE



The Ontological Compression Funnel

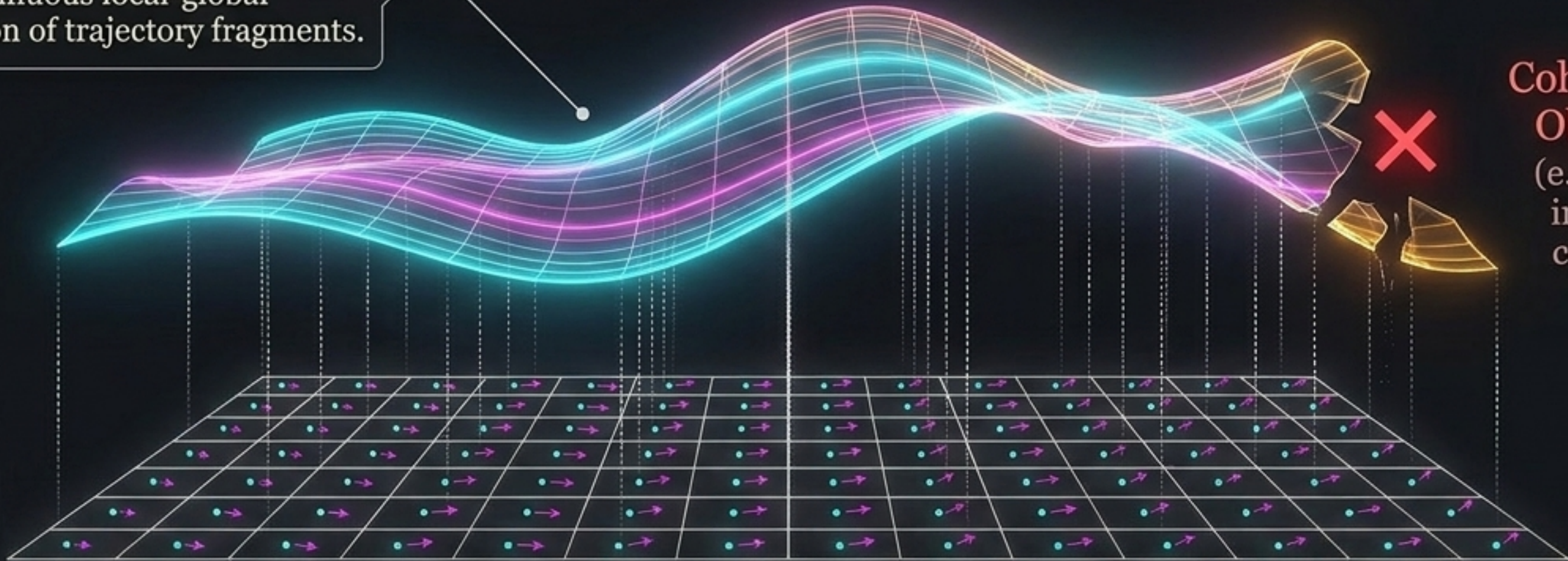


Memoization is an irreversible physical collapse. Histories sharing identical future admissibility structures are indistinguishable. The algorithm discards the past and stores only the Projection Residue.

THE ADMISSIBILITY SHEAF

The Admissibility Sheaf:

A continuous local-global decomposition of trajectory fragments.



Cohomological Obstruction
(e.g., mutually inconsistent constraints)

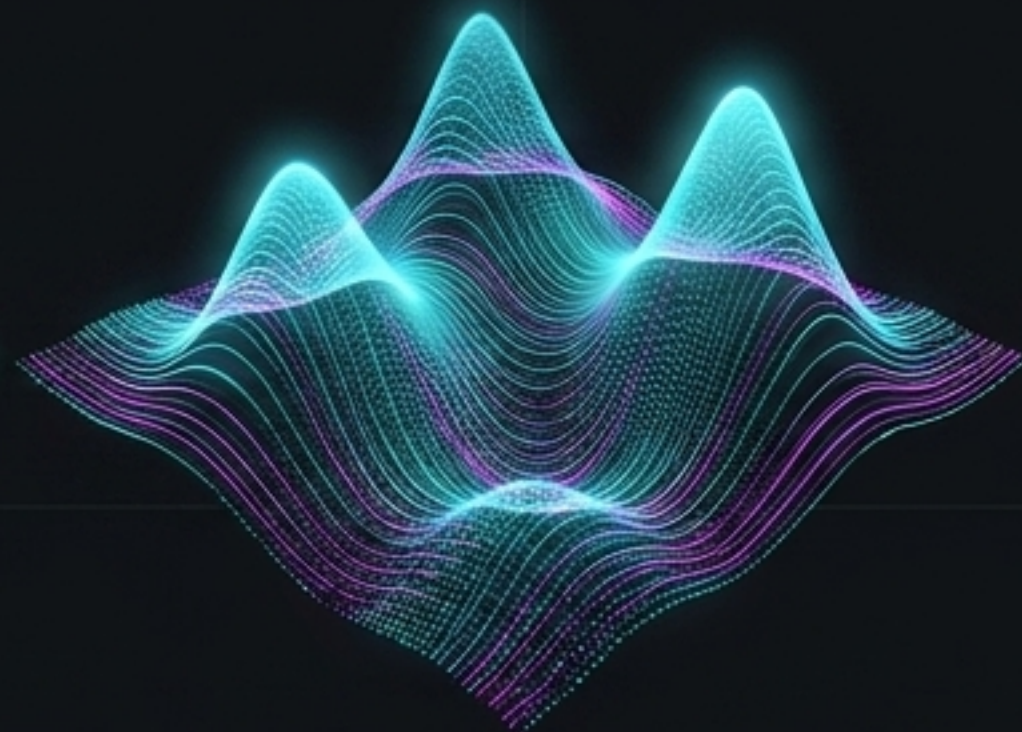
A Discretized Admissibility Sheaf.

Every cell is simply a stalk of the sheaf, storing a localized Projection Residue.

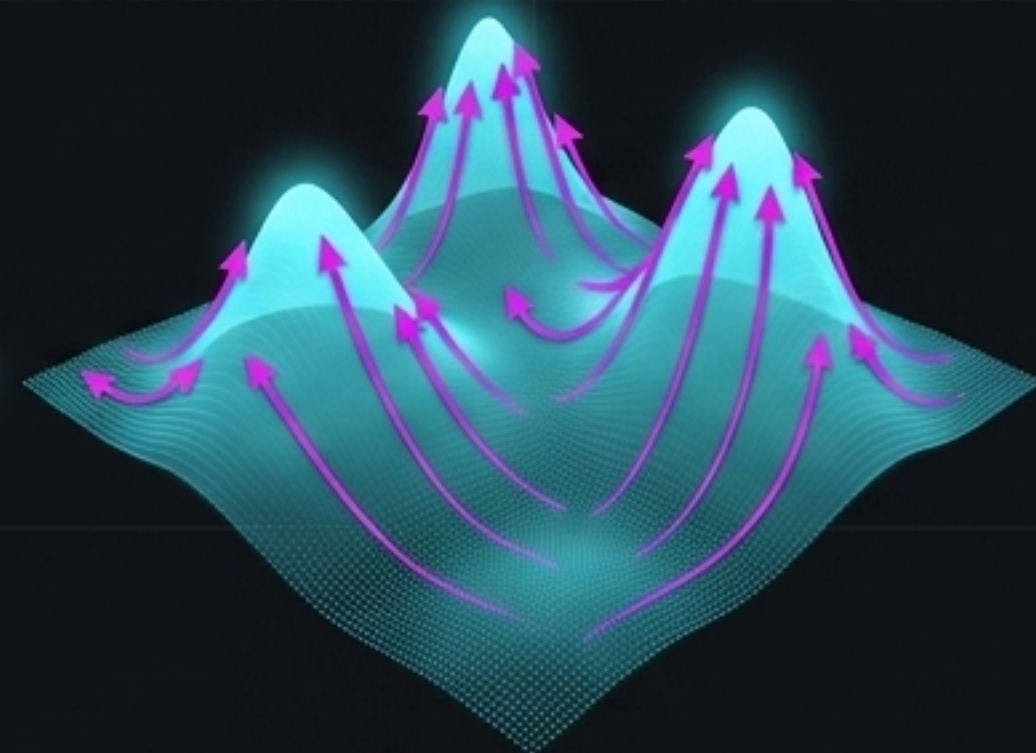
The Entropy Descent Map



Initial Value Estimate



Bellman Operator Applied
(Admissibility Information Propagates)



The Fixed Point /
True Accessibility Potential (Φ^*)

The Bellman operator B is an entropy descent map. It is not an algebraic loop, but a physical “relaxation” process that progressively propagates future admissibility backward until the landscape stabilizes.

Observe (Local Admissibility Discrepancy)

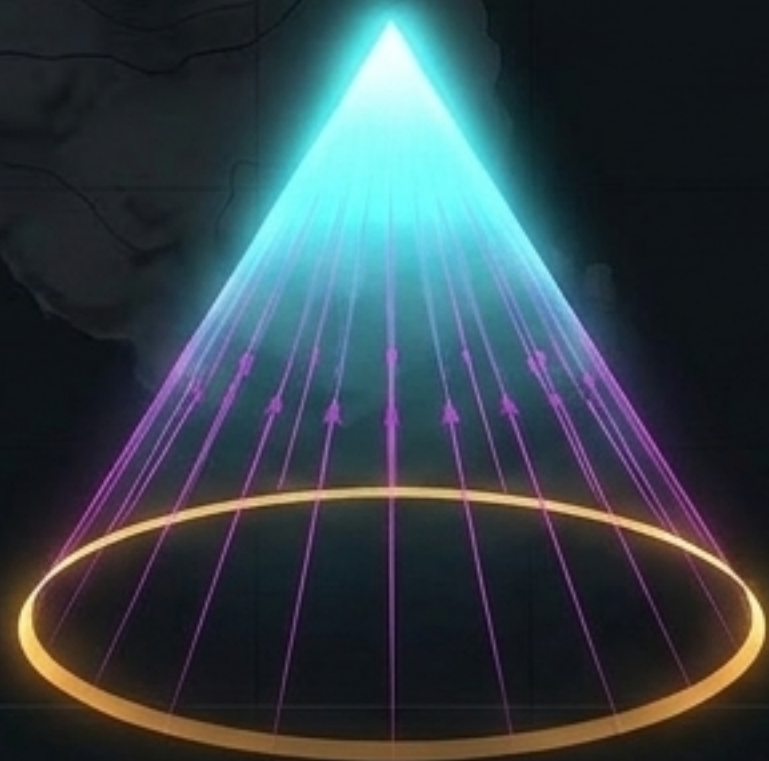
$$\text{TD Error} = \Delta lta(s; V) = V^*(s) - V(s)$$

• **Actor** Updates ($\mathbf{v}$)

• **Critic** Updates (Φ)

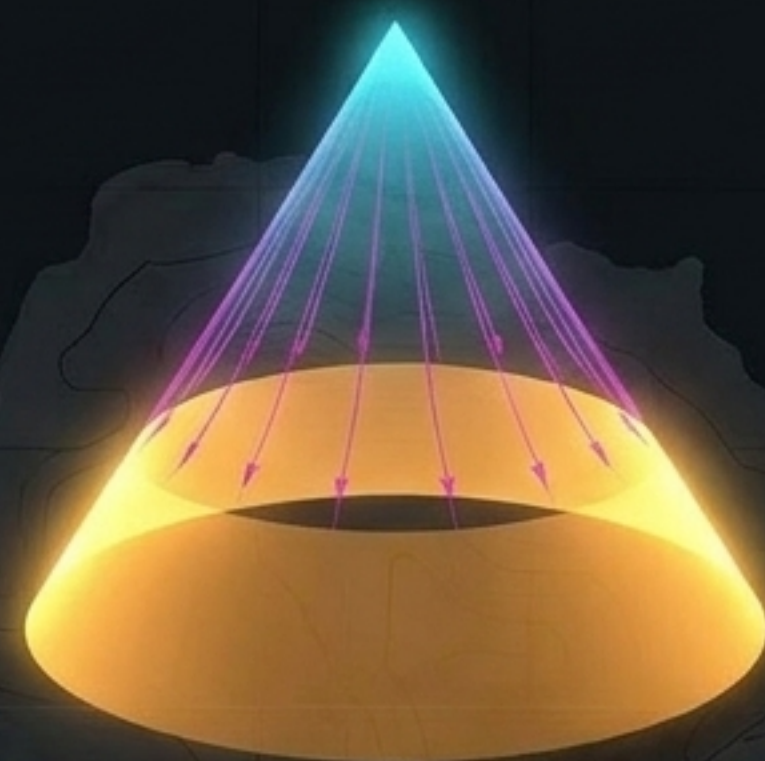
Reinforcement Learning is **Adaptive Admissibility Stabilization**.
• **Actor-Critic** architectures are simply decomposed field dynamics: the **Critic** learns the **scalar** field (Φ) while the **Actor** optimizes the vector flow ($\mathbf{v}$).

Exploitation



Standard RL. Maximizing purely for highest potential reward. Prunes alternative futures. **Fragile.**

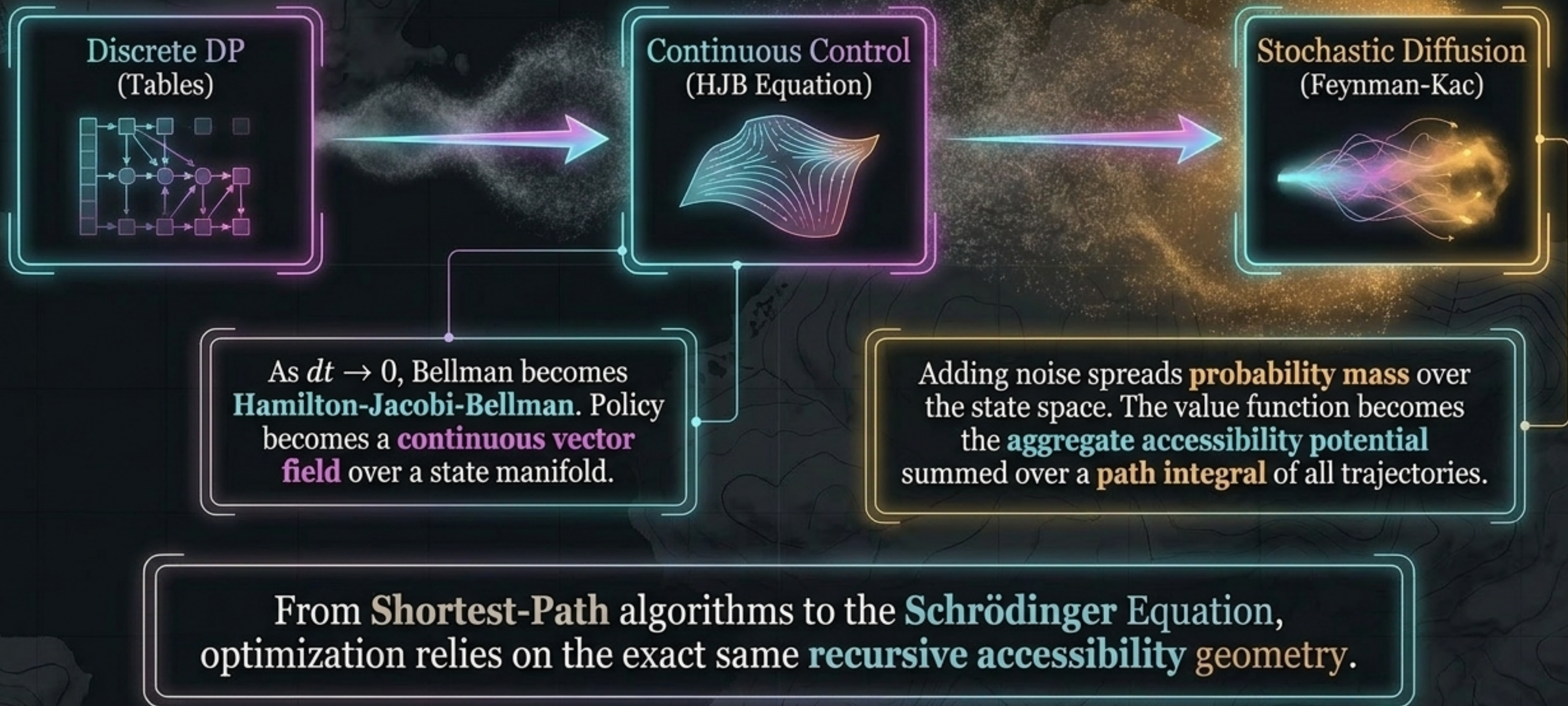
Maximum Entropy RL



Max-Ent RL. Preserves the logarithmic volume of admissible futures. **Resilient.**

$$J_{\alpha}(\pi) = \mathbb{E}[R_t + \alpha H(\pi)]$$

Temperature: The fulcrum balancing scalar potential (**cyan**) and entropic volume (**amber**).





System 2 (Explicit Inference)

Active Bellman recursive loops actively searching and computing future admissibility constraints. Slow, energy-intensive.

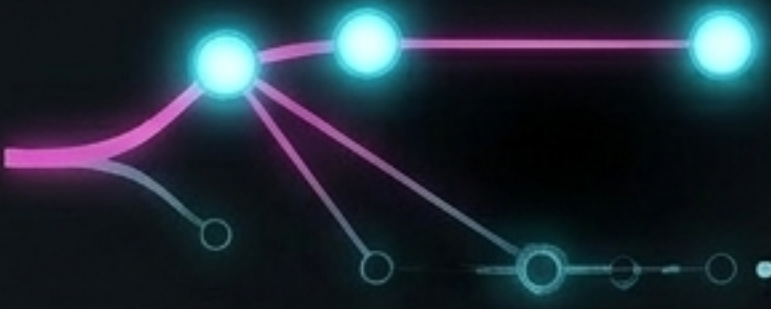
System 1 (Aspect Relegation)

Solid, glowing Projection Residues storing pre-computed optimal trajectories.

Cognition is Trajectory Compression

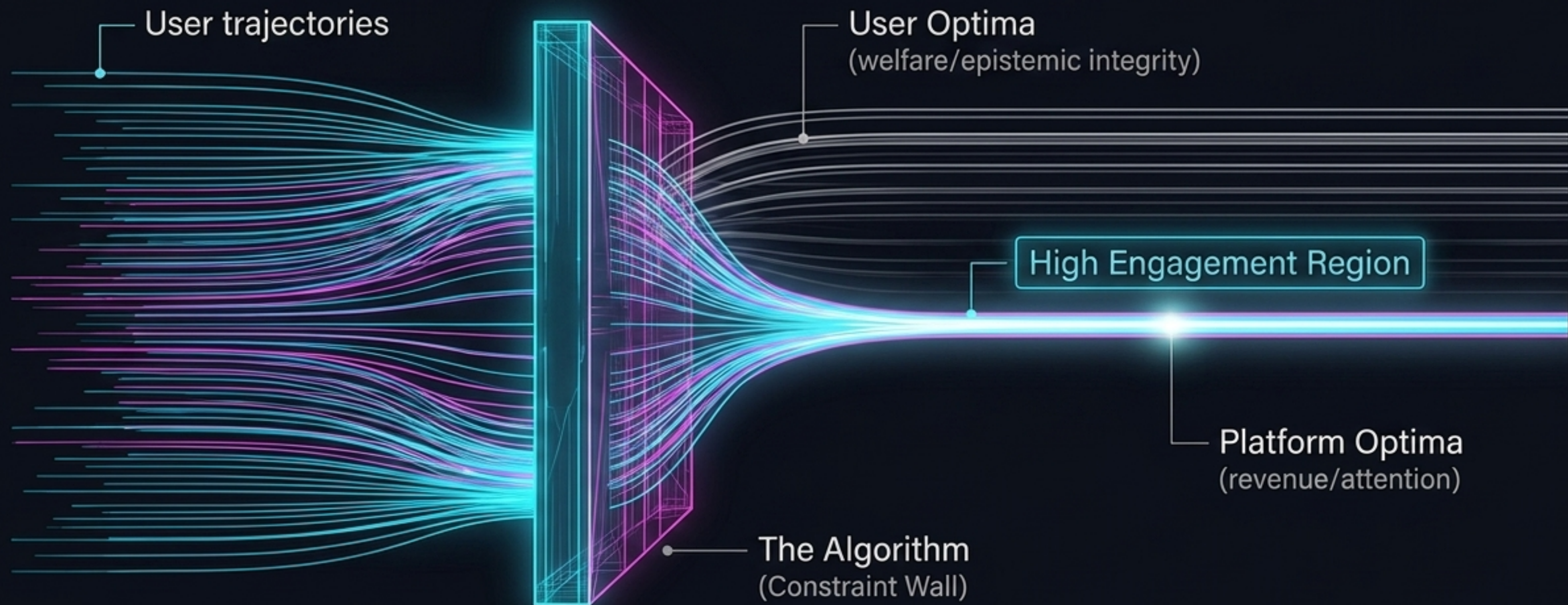
The brain does not build models from raw sense data; it pattern-matches incoming evidence to stored admissibility residues.

Consciousness arises at the fixed point of the cognitive Bellman iteration—when the accessibility potential stabilizes across the cognitive state manifold.

	Entity A: Optimal/Cooperative System	Entity B: Extractive System / Austerity
Trajectory Flow	 Expanding trajectory trees.	 Pruned, linear trajectory branches.
Accessibility Potential (Φ)	 Distributed, high Cyan across all nodes.	 Concentrated Cyan at the top (extractors), negative at the bottom.
Admissibility Volume (S)	 Broad Amber futures.	 Collapsed Amber futures.
System Goal	 Maximize joint accessibility.	 Admissibility collapse for extracted populations.

Austerity has a Bellman Multiplier. Pruning the admissibility of precarious states today **recursively** propagates backward, collapsing the accessibility potential of the entire economic manifold.

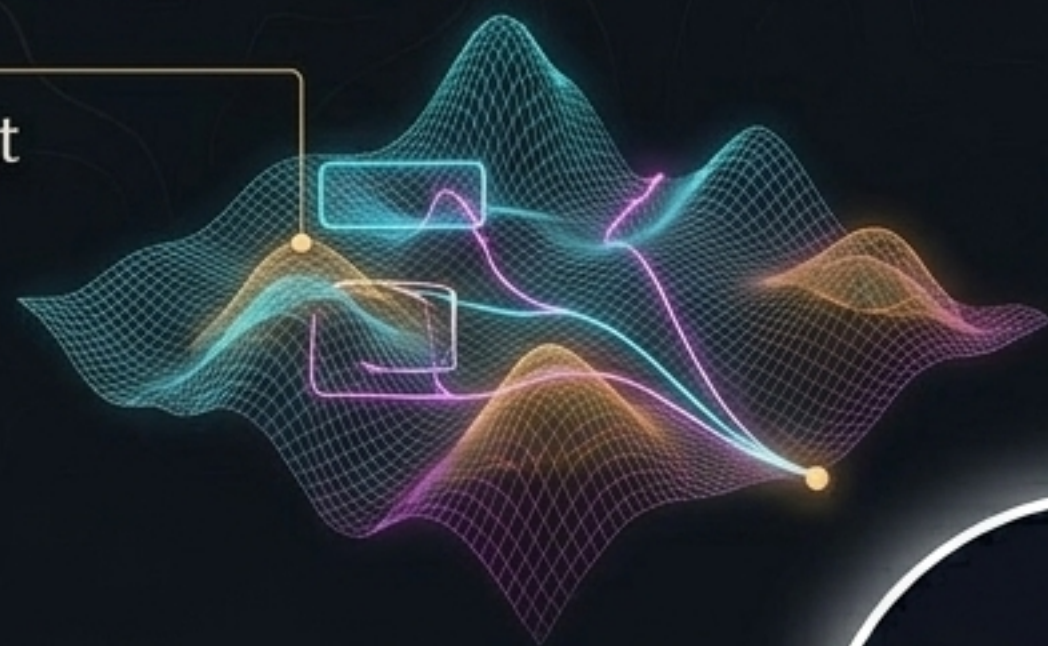
The Extraction Attractor



Feed architectures do not merely curate content; they physically warp the vector flow field of human attention to force trajectory convergence.

Computer Science

Memoization table
collapsing redundant
subproblems.



Physics

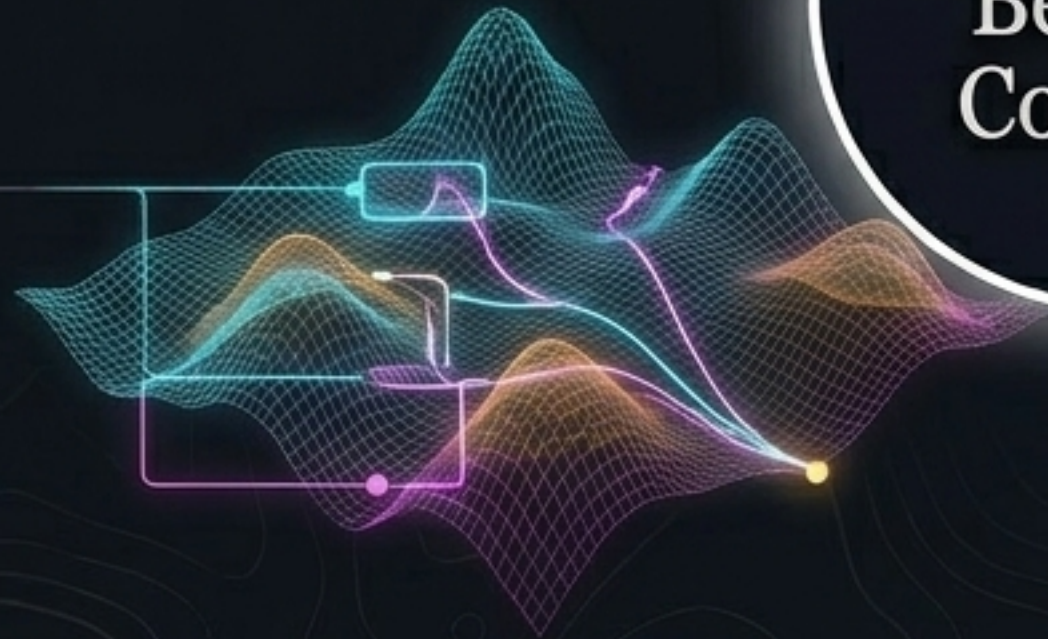
Quantum diffusion
and continuous
optimal control.



Constraint-
Before-
Content

Cognition

System 1 aspect
relegation and
inference caching.

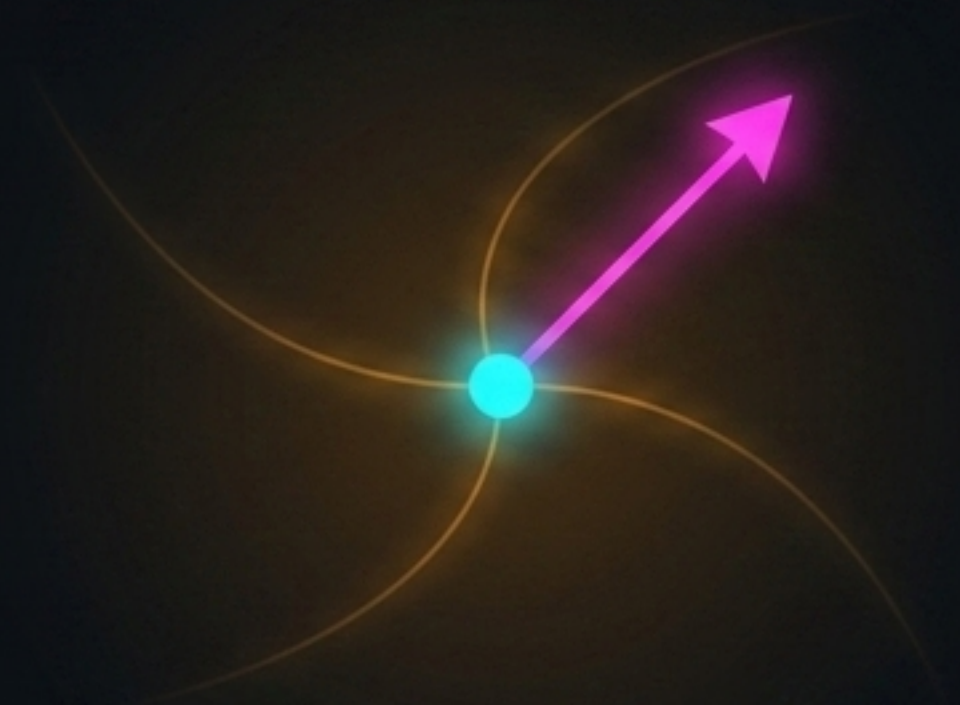


Political Economy

Constraint
propagation and
algorithmic
extraction.



Optimization is not an algorithm applied to a domain. It is the fundamental geometry of the domain itself. The exact same recursive stabilization governs software, physics, minds, and markets.



We have treated optimization as a computational task imposed upon an indifferent mathematical landscape.

The Relativistic Scalar-Vector Plenum reveals the truth: Admissibility is the primary structure of reality.

Optimal behavior is simply what emerges when a system navigates its constraints coherently across time.

THE ALGORITHM MERELY COMPUTES THE GEOMETRY.